For example, if the alpha forecast is that the stock will increase 5% over the next three days we have $\mu_{bp} = 500$ bp and $d = 3$.

We can further express our trading time t in terms of our trading strategy α as follows:

$$t = \frac{Shares}{ADV} \cdot \frac{1}{\alpha} \tag{11.20}$$

Then our alpha cost is:

$$Alpha\ Cost_{bp} = \frac{1}{2} \cdot \frac{\mu_{bp}}{d} \cdot \frac{Shares}{ADV} \cdot \frac{1}{\alpha} \tag{11.21}$$

Our simplified I-Star market impact model is:

$$MI_{bp} = a_1 \cdot \left(\frac{Shares}{ADV}\right)^{a_2} \cdot \sigma^{a_3} \cdot \alpha \tag{11.22}$$

In a properly structured alpha capture program the manager will seek to maximize the expected profit from this opportunity. This is determined as:

$$Max \quad \pi = \mu_{bp} - \left(\frac{\mu_{bp}}{2d} \cdot \frac{Shares}{ADV} \cdot \frac{1}{\alpha} + a_1 \cdot \left(\frac{Shares}{ADV}\right)^{a_2} \cdot \sigma^{a_3} \cdot \alpha\right) \tag{11.23}$$

The solution to the problem is:

$$\alpha^* = \sqrt{\frac{\mu_{bp}}{2d} \cdot \frac{Shares}{ADV} \cdot \left(a_1 \cdot \left(\frac{Shares}{ADV}\right)^{a_2} \cdot \sigma^{a_3}\right)^{-1}} \tag{11.24}$$

The maximum alpha capture opportunity is:

$$\pi^* = \mu_{bp} - \left(\frac{\mu_{bp}}{2d} \cdot \frac{Shares}{ADV} \cdot \frac{1}{\alpha^*} + a_1 \cdot \left(\frac{Shares}{ADV}\right)^{a_2} \cdot \sigma^{a_3} \cdot \alpha^*\right) \tag{11.25}$$

Example 6

A small cap stock is expected to increase 3% in the next 3 days. The next most attractive investment will increase 2% in the next three days. The manager wants to answer three questions:

1. How much alpha can a manager capture if the order size is 10% ADV?
2. How much can be invested in this stock before we begin to incur a loss?